Effat University
CS4082_ Machine Learning

Academic Research Poster

Health Risk Prediction Under Distribution Shift - Robust Machine Learning Across Heterogeneous Patient Populations

CS4082 – MACHINE LEARNING, EFFAT UNIVERSITY · SPRING 2026

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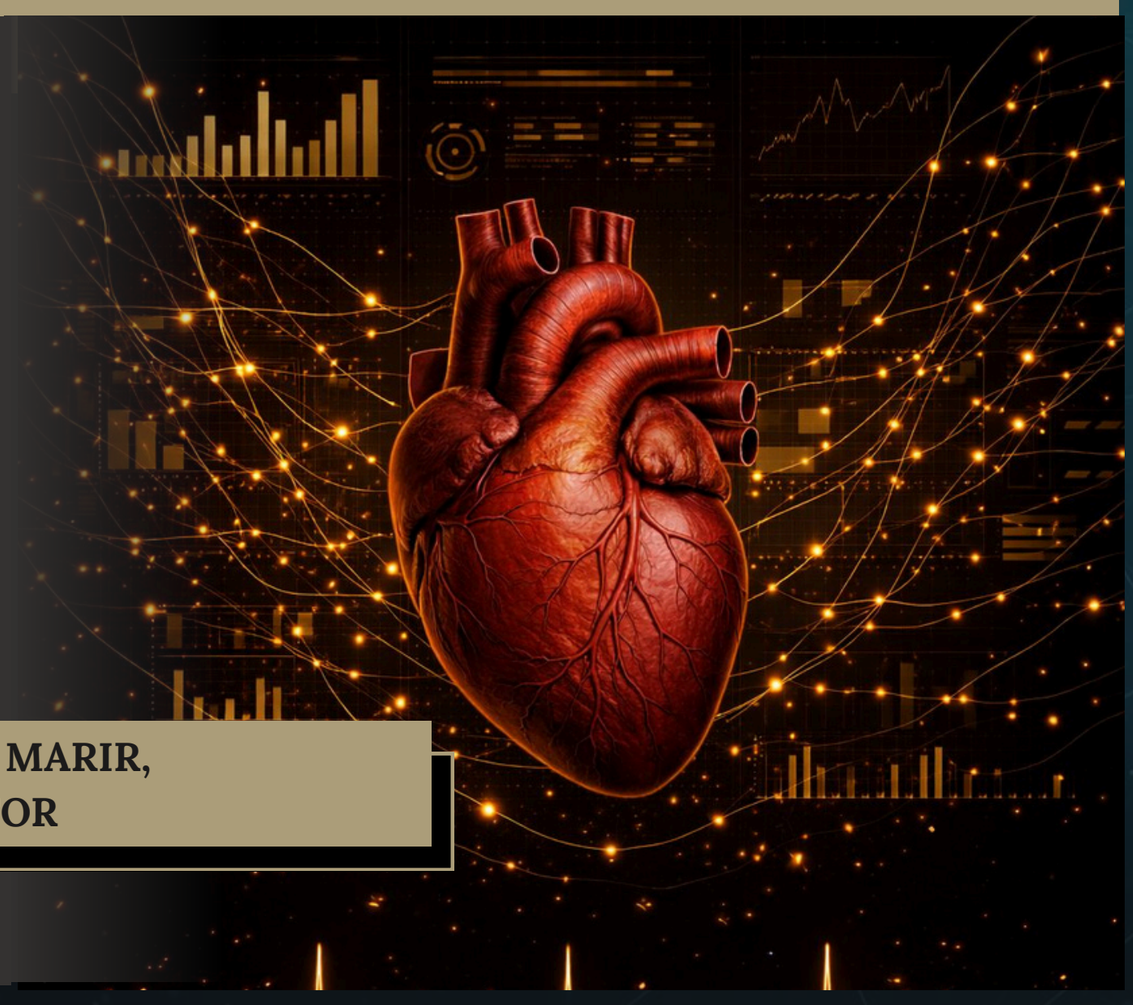

Introduction

- *Distribution shift: deployment data differs from training data*
- *Examples: Different hospitals, populations, disease prevalence*
- *Models may perform well in development but fail in deployment*

RESEARCH QUESTION

Can models trained on one cardiovascular dataset **remain reliable under distribution shift**, and which mitigation strategy improves performance the most?

SOURCE DOMAIN: UCI Heart Disease - 918 patients, 11 clinical features, 55.3% positive rate. Used for training & validation.

TARGET DOMAIN: Framingham Heart Study - 4,240 subjects, 15 features, 15.2% positive rate.

EVIDENCE OF SHIFT: Severe label shift 55.3%→15.2%, Covariate shift confirmed (KS test p<1e-10), Multi-dimensional shift.

METHODOLOGY PIPELINE

Data Preparation
In-Domain Baselines
Class Weighting & SMOTE
Threshold Tuning
Final Comparison

Exploratory Data Analysis
Baseline OOD Evaluation
Hyperparameter Tuning
Pooled Training
Results Analysis

KEY RESULTS

In-Domain	**0.9036**
Best Baseline OOD	**0.6848**

BEST STRATEGY

Pooled Training: ROC-AUC 0.7000

FINAL TAKEAWAY

More diverse training data improves robustness more than isolated model tuning.

CONCLUSION

Distribution shift harms reliability · Diverse training data improves robustness most

LIMITATIONS

- ONLY 4 SHARED PREDICTORS
- DIFFERENT LABEL DEFINITIONS
- EVALUATION SIZE DIFFERENCES

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